Practice Sheet 1.5: Basics of Python Programming

- **Easy:** for, while
- **Medium:** nested loops, counters
- **Hard:** nested loops + math logic + conditionals
  Perfect progression for **1st–2nd year students and placement preparation**.

□ Python Looping Statements – Problem Statements

□ EASY LEVEL (10 Problems)

1. Print Natural Numbers

Problem Statement:
Write a Python program to print all natural numbers from 1 to N using a loop.

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2. Sum of First N Numbers

Problem Statement:
Given a number N, calculate the sum of the first N natural numbers using a loop.

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3. Even Numbers in a Range

Problem Statement:
Given two integers start and end, print all even numbers between them (inclusive) using a loop.

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4. Multiplication Table

Problem Statement:
Write a program to display the multiplication table of a given number N up to 10 using a loop.

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5. Count Digits in a Number

Problem Statement:
Given an integer N, count the number of digits in it using a loop.

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6. Reverse a Number

Problem Statement:
Write a program to reverse a given number using a loop.

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7. Factorial of a Number

Problem Statement:
Given a number N, find its factorial using a loop.

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8. Print Characters of a String

Problem Statement:
Given a string S, print each character of the string on a new line using a loop.

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9. Sum of Digits

Problem Statement:
Write a program to find the sum of digits of a given number using a loop.

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10. Count Vowels in a String

Problem Statement:
Given a string, count the number of vowels present in it using a loop.

MEDIUM LEVEL (10 Problems)

11. Prime Number Check

Problem Statement:
Write a program to check whether a given number is prime using a loop.

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12. Print All Prime Numbers in a Range

Problem Statement:
Given two numbers L and R, print all prime numbers between them using nested loops.

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13. Fibonacci Series

Problem Statement:
Given a number N, print the first N terms of the Fibonacci series using a loop.

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14. Armstrong Number

Problem Statement:
Write a program to check whether a given number is an Armstrong number using a loop.

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15. Palindrome Number

Problem Statement:
Given a number, check whether it is a palindrome using a loop.

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16. Count Frequency of Digits

Problem Statement:
Given a number, count the frequency of each digit using loops.

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17. Find GCD of Two Numbers

Problem Statement:
Write a program to find the Greatest Common Divisor (GCD) of two numbers using a loop.

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18. Pattern Printing – Right Triangle

Problem Statement:
Using loops, print a right-angled triangle pattern of * of height N.

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19. Sum of Prime Digits

Problem Statement:
Given a number, find the sum of all prime digits present in it using a loop.

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20. Perfect Number Check

Problem Statement:
Write a program to check whether a given number is a perfect number using a loop.

HARD LEVEL (10 Problems)

21. Strong Number

Problem Statement:
A number is called a **Strong Number** if the sum of the factorials of its digits is equal to the number itself.
Write a program to check whether a given number is a Strong Number using loops.

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22. LCM of Two Numbers

Problem Statement:
Write a program to find the Least Common Multiple (LCM) of two numbers using loops.

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23. Print Prime Factors

Problem Statement:
Given a number, print all its prime factors using loops.

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24. Number Pattern – Pyramid

Problem Statement:
Using nested loops, print a pyramid number pattern for a given height N.

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25. Count Special Characters

Problem Statement:
Given a string, count the number of alphabets, digits, and special characters using loops.

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26. Find Second Largest Number

Problem Statement:
Given a list of integers, find the second largest number using loops (without using built-in sorting).

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27. Sum of Series

Problem Statement:
Calculate the sum of the series:
1 + 11 + 111 + ... up to N terms using loops.

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28. Binary to Decimal Conversion

Problem Statement:
Given a binary number, convert it into its decimal equivalent using loops.

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29. Matrix Row and Column Sum

Problem Statement:
Given a matrix of size M × N, find the sum of each row and each column using nested loops.

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30. Pascal’s Triangle

Problem Statement:
Write a program to print Pascal’s Triangle up to N rows using nested loops.